Chuanpeng Hou PhD

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Education

2024 **PhD degree, Peking University** in Space Physics

2022 – 2024 **Visiting PhD student, IRAP, Toulouse, France** in Space Physics

2019 **Bachelor's degree, Peking University** in Space Physics

Fellowships, Awards and Scholarships

Fellowships

2025 **Alexander von Humboldt Research Fellowship**, Alexander von Humboldt-Stiftung

Awards

2024 **Best Student Paper Awards**, National Planetary Science Conference.
2023 **Merit Student**, Peking University.
2022 **Best Student Paper Awards**, Chinese Geoscience Union (CGU).
Best Student Poster Awards, Asia Oceania Geosciences Society (AOGS).
2021 **Graduate Award for Scientific Research**, Peking University.

Scholarships

2023 **Peking University President's Scholarship**, Peking University.
2022 **Chinese Government Scholarship**, China Scholarship Council.

Conference Organization

2026 **Co-convener**, Session ST29: Linking Plasma Dynamics in the Lower Solar Atmosphere to Magnetic Activity, AOGS Annual Meeting

Conference Presentations

Invited talks

2026 The Contribution of Turbulence to Solar Wind Heating in the Inner Heliosphere, 2nd Plasma Astrophysics School and Workshop (PASW26).

Research Publications

Journal Articles (†, co-first author)

1. **C. Hou**, J. He, A. P. Rouillard, C. Li, S. Rao, X. Zhu, Z. Wu, D. Verscharen, Z. Huang, L. Yang, et al., "Non-adiabatic entropy increase in the solar wind governed by velocity spikes", Nature Astronomy, 1–11 (2026).
2. **C. Hou**, H. Yan, and S. Zhao, "Compressible turbulence as a source of particle beams and ion bernstein waves in collisionless plasmas", The Astrophysical Journal **1004**, 64 (2026).
3. **C. Hou**, H. Yan, S. Zhao, and P. Pavaskar, "Energy cascade and damping in fast-mode compressible turbulence", The Astrophysical Journal Letters **992**, L28 (2025).
4. **C. Hou**†, B. Gannouni†, A. P. Rouillard, J. He, and V. Réville, "Fine-scale activity driven by magnetic reconnection within coronal microjets", Astronomy & Astrophysics **697**, A67 (2025).
5. **C. Hou**, J. He, D. Duan, Z. Wu, Y. Chen, D. Verscharen, A. P. Rouillard, H. Li, L. Yang, and S. D. Bale, "The origin of interplanetary switchbacks in reconnection at chromospheric network boundaries", Nature Astronomy **8**, 1246–1256 (2024).
6. **C. Hou**, A. P. Rouillard, J. He, B. Gannouni, V. Réville, P. Louarn, A. Fedorov, L. Přech, C. J. Owen, D. Verscharen, R. D'Amicis, L. Sorriso-Valvo, N. Fargette, J. Coburn, V. Génot, J. M. Raines, R. Bruno, S. Livi, B. Lavraud, N. André, G. Fruit, R. Kieokaew, I. Plotnikov, E. Penou, A. Barthe, D. Kataria, M. Berthomier, F. Allegrini, V. Fortunato, G. Mele, and T. Horbury, "Connecting solar wind velocity spikes measured by solar orbiter and coronal brightenings observed by sdo", The Astrophysical Journal Letters **968**, L28 (2024).
7. **C. Hou**†, X. Zhu†, R. Zhuo†, J. He, D. Verscharen, and D. Duan, "Nature, generation, and dissipation of alfvénic kinks/switchbacks observed by parker solar probe and wind", The Astrophysical Journal **950**, 157 (2023).
8. **C. Hou**, J. He, D. Duan, X. Zhu, W. Li, D. Verscharen, T. Liu, and T. Wang, "Efficient energy conversion through vortex arrays in the turbulent magnetosheath", The Astrophysical Journal **946**, 13 (2023).
9. **C. Hou**, J. He, X. Zhu, and Y. Wang, "Contribution of magnetic reconnection events to energy dissipation in space plasma turbulence", The Astrophysical Journal **908**, 237 (2021).
10. **C. Hou**, J. He, L. Zhang, Y. Wang, and D. Duan, "Dynamics of the charged particles released from a sun-grazing comet in the solar corona", Earth and Planetary Physics **5**, 232–238 (2021).